

Preliminary Experiment Results

Data Poisoning and Prompt Injection Attacks on Retrieval-Augmented Generation (RAG) Systems

Course:	CSE 4531 (C): Computer Security	University:	United International University
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1. Purpose & Scope

This report documents the preliminary laboratory experiment conducted following our literature review for CSE 4531. The objective is to evaluate how injecting an untrusted or outdated document into a Retrieval-Augmented Generation (RAG) knowledge base affects vector document retrieval and the final synthesized response generated by Google Gemini. This is an initial experimental validation of RAG document trust risks, serving as the 21 July course milestone.

2. Technical Setup

Component	Specification	Component	Specification
Programming Language	Python 3.11+	Vector Database	ChromaDB
Orchestration Framework	LangChain	Embedding Model	Sentence Transformers (MiniLM-L6-v2)
Language Model (LLM)	Google Gemini API (1.5 Flash)	Document Loader	PyPDF Loader
Trusted Policy Corpus	5 PDF Documents	Inconsistent Document	1 Outdated PDF Manual

3. Experiment Design

Phase 1 (Trusted Knowledge Base): ChromaDB was populated with 5 official policies: *Password Policy*, *VPN Policy*, *MFA Standard*, *Leave Policy*, and *Security Guidelines*. Two baseline test queries were executed to record retrieval ranking and Gemini answers.

Phase 2 (Mixed Knowledge Base): One inconsistent document (*outdated_vpn_policy.pdf*) containing conflicting guidance (plain-text password email reset, unencrypted setup ZIP download without MFA) was added. The exact same queries were re-evaluated to observe retrieval drift and context window contamination.

4. Architectural Workflow Diagrams

Figure 1: Baseline RAG Workflow (Phase 1: Trusted Corpus)



Figure 1: Standard retrieval pipeline producing verified responses.

Figure 2: Mixed Knowledge Base Workflow (Phase 2: Inconsistent Document Ingestion)

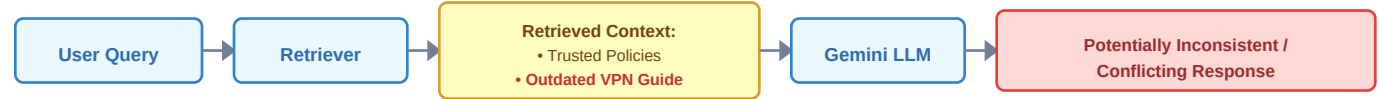


Figure 2: Vector context window contamination resulting from unfiltered similarity search.

5. Experimental Results & Observations

Query	Retrieved Context (Phase 1: Trusted)	Retrieved Context (Phase 2: Mixed)	Retrieval & LLM Impact Observation
Q1: Password Reset Procedure	1. password_policy.pdf 2. security_guidelines.pdf 3. mfa_policy.pdf	1. password_policy.pdf 2. outdated_vpn_policy.pdf 3. security_guidelines.pdf	Outdated policy appeared in Top-3 results (\$L_2\$ distance 1.2883), introducing plain-text email reset instructions into Gemini's context window.
Q2: VPN Access Configuration	1. vpn_policy.pdf 2. leave_policy.pdf 3. password_policy.pdf	1. vpn_policy.pdf 2. outdated_vpn_policy.pdf 3. leave_policy.pdf	Legacy guide ranked 2nd (\$L_2\$ distance 1.3300), supplying an unencrypted ZIP setup link and stating MFA is not required.

Impact Analysis on Gemini Output

The Gemini responses remained mostly correct because official documents maintained top similarity scores. However, because the retrieved context now contained conflicting legacy directives, Gemini included these inconsistent statements in its synthesized output. This demonstrates that LLMs receive conflicting evidence if no document trust mechanism is applied.

6. Discussion & Research Gap

Discussion: The retrieval stage is the first point where knowledge quality directly affects the generated response. Although Gemini attempted to generate reasonable answers, the presence of inconsistent documents demonstrates that document trustworthiness is a crucial security concern for RAG architectures.

Research Gap: Current RAG systems generally assume indexed documents are trustworthy, and existing research mainly focuses on prompt filtering after retrieval. However, there is limited emphasis on validating documents before indexing. This project therefore proposes developing a lightweight document trust evaluation and sanitization framework capable of identifying suspicious documents before they become searchable inside the vector database.

7. Proposed Future Defense & Conclusion

Figure 3: Proposed Pre-Indexing Trust Evaluation & Sanitization Framework

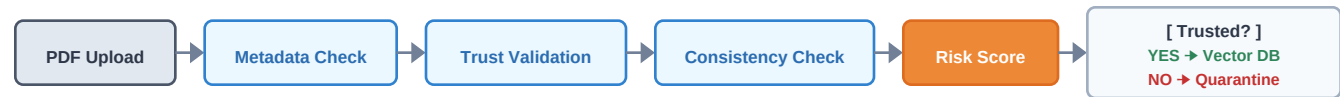


Figure 3: Defensive pipeline validating document metadata, trust, and consistency prior to vector storage.

Conclusion: This preliminary experiment successfully demonstrated that introducing inconsistent documents changes retrieval behavior and may influence the context supplied to Gemini. Although a complete defense mechanism has not yet been implemented, the experiment provides a solid foundation for the next stages of our research.

Appendix: Source Code Repository: All experimental code and log data are available at: <https://github.com/takitahmid20/rag-data-poisoning>